

1 Question (15 points)

Write a method named `aBeforeB` which takes a string parameter and returns the number of 'a' characters preceding a 'b' character in the string *s*.

A call to `aBeforeB` by parameter "acabab" should return 2.

2 Question (20 points)

Write a method named **yardSale** that takes an initial amount of money and the number of purchase trials as its parameter, and returns the amount of money the user ends up with after a series of purchases.

Your method should ask the user for a series of potential purchases specified by a unit price per single item and the quantity of that item. The user buys everything that does not exceed his current amount of money.

A purchase is considered a **good deal** if the unit price is under 10 TL, **a message should be printed** when such a purchase is made. In addition, your method should print the **total quantity of items purchased** and the total cost of the **most expensive** item purchased. With an initial money of 50.75 TL and 4 purchase trials:

Price? 8.75

Quantity? 2

What a deal! I'll buy it.

Remaining money: 33.25

Price? 11.50

Quantity? 3

Remaining money: 33.25

Price? 5.10

Quantity? 5

What a deal! I'll buy it.

Remaining money: 7.75

Price? 0.75

Quantity? 8

What a deal! I'll buy it.

Remaining money: 1.75

Total quantities purchased: 15

Most expensive purchase: 25.5

3 Question (15 points)

Write a method name **thirdSmallestDivisor** which takes **two integer parameters** and **returns the third smallest common divisor** of those integers. If the integers do not have a third smallest common divisor the method will return -1.

Write a main method such that inside the main method, read two integers from the user and using the method `thirdSmallestDivisor` find and print the third smallest common divisor.

Enter two integers:

12

8

4

4 Question (15 points)

A palindrome is a sequence of characters that reads the same backward as forward. For example, each of the following four-digit integers is a palindrome: 1221, 5555, 4554 and 8118. Write a program that reads in a four-digit integer and determines whether it is a palindrome. If the number is not four digits long, display an error message and exit the program. Otherwise, check if it's a palindrome or not.

Enter number: 1441

The number is palindrome

5 Question (15 points)

In this question you will read an integer n from the user and then read n more integers. You should find the smallest (min) and the largest number (max) entered by the user and calculate min^{max} without using pow method from the Math library. You are given the main method. **You will write a method named calculatePower which takes a Scanner parameter.**

1. First, you will ask the user how many numbers she wants to enter. You should do this inside the method.
2. Then user will enter the numbers as shown in the example. You may assume that the user will enter integers between 0 and 100.
3. calculatePower method will **print the minimum and the maximum of the grades** entered by the user.
4. calculatePower method will calculate and **return the minimum number raised to the power maximum.**

How many numbers:

4

4 2 6 3

Minimum: 2

Maximum: 6

Answer: 64.0

6 Question (20 points)

Write a program which simulates the trip of a car.

1. The traffic lights are green at time 0.
2. The traffic lights are green or red for the same time interval.
3. The car starts from an initial point and travels until the first traffic light.
4. If the first light is green, the car continues, otherwise it waits for the green signal.
5. Afterwards, the car continues and travels until the second traffic light.
6. If the second light is green, the car continues, otherwise it waits for the green signal.

The user enters (i) the speed of the car in m/s, (ii) the length of the road (in meters) from initial point to the first light, (iii) time interval for the first light (in seconds), (iv) the length of the road (in meters) from the first light to the second light, (v) time interval for the second light (in seconds). The program will output the time required to finish the trip.

Speed: 4

1. Road Length: 100

1. Time Interval: 20

2. Road Length: 100

2. Time Interval: 20

Simulation Time: 80